

APJ Abdul Kalam Technological University

MASTER OF ARCHITECTURE
(Environmental Design)

CURRICULUM

JUNE 2025

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Master of Architecture (Environmental Design) – Curriculum

FIRST SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
251TED001	Behaviour in Built Environment	2-1-0	3	40	60	2.5	A
251TED002	Fundamentals of Material Properties and Construction Techniques	2-1-0	3	40	60	2.5	B
251TED003	Building Physics I	3-2-0	4	40	60	2.5	C
251LED001	Environmental Design Lab I	0-4-0	2	40	-	-	-
251SED001	Environmental Design Studio I	0-0-9	9	200	100		V*
	TOTAL	24	21	360	280		

*Jury Examination

SECOND SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
252TED001	Renewable Energy and Building Systems	2-1-0	3	40	60	2.5	A
252TED002	Building Physics II	2-1-0	3	40	60	2.5	B
252RED001	Research Methodology and Seminar	1-2-0	2	40	60	2.5	S
252EED***	Interdisciplinary elective	2-1-0	3	40	60	2.5	T
252EAR***	Department Elective	1-1-0	2	40	60	2.5	U
252LED001	Environmental Design Lab II	0-4-0	2	40	-	-	-
252SED001	Environmental Design Studio II	0-0-8	8	200	100		V*
	TOTAL	26	23	440	400		

*Jury Examination

Interdisciplinary Electives: (Offered by Environmental Design Stream for students in second semester of other branches)	
252EED001	Green Building Rating Systems
252EED002	Sustainable Architecture

Department Electives: (Offered to students of all M Arch and M Plan programmes)	
252EAR001	Urban Conservation and Regeneration
252EAR002	Environmental Impact Assessment
252EAR003	Urban Sociology
252EAR004	Contemporary Pedagogy in Architectural Education

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THIRD SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
253TED001	Landscape Ecology and Disaster Resilience	2-1-0	3	40	60	2.5	A
253TED002	Advanced Material Performance	1-1-0	2	40	60	2.5	B
253MED001	Elective I (MOOC)	-	2	-	-	-	-
253PED001	Dissertation	0-6-0	3	100	100	-	U
253LED001	Environmental Design Lab III	0-4-0	2	40	-	-	-
253SED001	Environmental Design Studio III	0-0-9	9	200	100	-	V*
253IAR001	Internship	-	2	100	-	-	-
	TOTAL	24	23	520	320		

*Jury Examination

Note:

Elective I shall be any MOOC as specified in the syllabus. The course could be done during the first, second or third semester so that the passing grade should be available by the third semester.

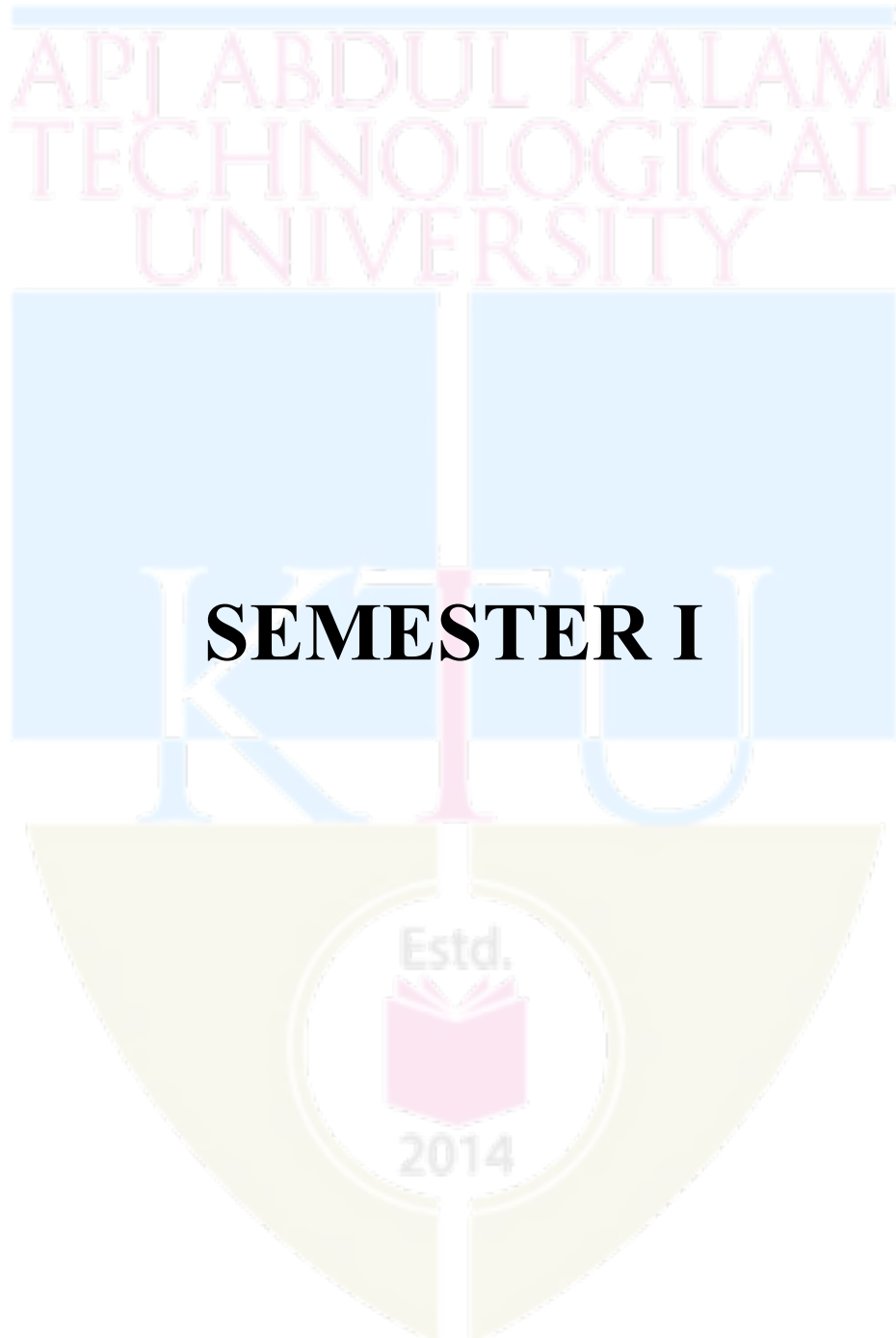
FOURTH SEMESTER

Course Code	Course	L-T-S	Credits	Internal Marks	End Semester Exam		Exam Slot
					Marks	Duration (Hours)	
254TED001	Environmental Economics and Management	2-1-0	3	40	60	2.5	A
254EED***	Elective II	1-1-0	2	40	60	2.5	S
254PED001	Environmental Design Thesis	0-0-21	9	350	150	-	V*
	Thesis Report	-	3	-	-	-	-
	TOTAL	26	17	430	270		

*Jury Examination

Stream Electives:	
254EED001	Circular Economy and Sustainable development
254EED002	Environmental Management
254EED003	Urban climate and Eco-cities
254EED004	Energy and Environmental Data Analytics





SEMESTER I

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TED001	Behaviour in Built Environment	2-1-0	3	2025
Course Objectives	This course aims to introduce students to key concepts and psychological theories that explain the relationship between human behaviour and the built environment. It examines theoretical frameworks and behavioural models relevant to architecture and urban design, enabling students to analyze behavioural responses across public, institutional, and residential spaces. Students will also learn to apply appraisal and evaluation methods to assess user experience and integrate behavioural wellness principles and healthy building strategies into architectural design.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Conceptualize and interpret human responses to built environments through key behavioural concepts	Understanding, Applying		
CO 2	Apply environmental-behavioural theories to evaluate and predict patterns in architectural and urban contexts.	Applying, Analyzing		
CO 3	Critically analyze built environments to propose responsive and inclusive design solutions based on user behaviour.	Analyzing, Creating		
CO 4	Assess the effectiveness of built spaces using behavioural appraisal tools and environmental evaluation methods.	Evaluating		
CO 5	Integrate healthy building principles to support physical, mental, and social well-being in design.	Applying, Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				

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End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Introduction to Behaviour and the Built Environment – • Introduction, Definitions, and Scope • Maslow's Hierarchy of Human needs, Gestalt Principles, Environmental Psychology Issues • Proxemics, territoriality, personal space, crowding, and density. • Place attachment, place identity, and place dependence. • Environmental perception and cognition. • Sense of place and placemaking principles. • Wayfinding behaviour and spatial cognition. • SDGs and Behaviour: SDG 3, 11, and 13 and their relation to user behaviour and design	CO1	9
II	Theoretical Framework and Behavioural Models – • Stimulus-Response theory and Behaviourism. • Environmental Habituation, Environmental Numbness • Prospect-Refuge theory, Routine Activity Theory, Attention-Restoration Theory, Stress Reduction Theory. • Defensible Space and CPTED principles. • Crime Opportunity Theory, Broken Windows Theory. • Social Learning theory and Observational learning in built environments. • Neuro-Architecture	CO2	9
III	Application Areas – Behavior in Urban and Architectural Contexts – • Behaviour in public spaces: Parks, streets, transport nodes • Institutional spaces: Schools, hospitals, workplaces. • Residential behaviour: Privacy, social interaction, home territories. • Inclusive environments and user diversity (age, gender, accessibility).	CO3	10

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	• Temporary vs. permanent spaces: Festivals, exhibitions, shelters.		
IV	Appraisals & Assessments – Environmental Behaviour Studies - • Types of appraisals and environmental assessments • Public Space Evaluations (Urban, transit settings, etc) • Private spaces Evaluations (Institutional, residential, etc.)	CO4	10
V	Healthy Buildings and Behavioural Wellness – • Concept of Healthy Buildings • Biophilic design and psychological restoration. • Sick Building Syndrome • Stress, cognition, and design-induced emotional responses. • WELL Building Standard and behavioural dimensions. • Standards and rating systems on healthy buildings	CO5	10

Reference Books/ Readings:

1. Altman, I. (1975). *The environment and social behavior: Privacy, personal space, territory, and crowding*. Brooks/Cole.
2. Hall, E. T. (1966). *The hidden dimension*. Doubleday.
3. Canter, D. (1977). *The psychology of place*. Architectural Press.
4. Relph, E. (1976). *Place and placelessness*. Pion.
5. Gifford, R. (2013). *Environmental psychology: Principles and practice* (5th ed.). Optimal Books.
6. Lang, J. (1987). *Creating architectural theory: The role of the behavioral sciences in environmental design*. Van Nostrand Reinhold.
7. Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.
8. Appleton, J. (1975). *The experience of landscape*. Wiley.
9. Newman, O. (1972). *Defensible space: Crime prevention through urban design*. Macmillan.
10. Zeisel, J. (2006). *Inquiry by design: Environment/behavior/neuroscience in architecture, interiors, landscape, and planning*. W. W. Norton.
11. Groat, L., & Wang, D. (2013). *Architectural research methods* (2nd ed.). Wiley.
12. Denscombe, M. (2017). *The good research guide* (6th ed.). Open University Press.
13. Flick, U. (2018). *An introduction to qualitative research* (6th ed.). SAGE.
14. Gehl, J. (2011). *Life between buildings: Using public space*. Island Press.
15. Madanipour, A. (1996). *Design of urban space: An inquiry into a socio-spatial process*. Wiley.
16. Lynch, K. (1960). *The image of the city*. MIT Press.
17. Francis, M., & Marcus, C. C. (1998). *People places: Design guidelines for urban open space*. Wiley.
18. Allen, J. G., & Macomber, J. D. (2020). *Healthy buildings: How indoor spaces drive performance and productivity*. Harvard University Press.
19. Heerwagen, J., Heschong, L., & Neapolitan, R. (2008). *Biophilic design: The theory, science and practice of bringing buildings to life*. Wiley.
20. Heschong, L. (1979). *Thermal delight in architecture*. MIT Press.

Standards & Reports:

1. WELL Building Standard v2 (International WELL Building Institute)
2. Harvard T.H. Chan School of Public Health - 9 Foundations of a Healthy Building



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TED002	Fundamentals of Material Properties and Construction Techniques	2-1-0	3	2025
Course Objectives	The course intends to create an understanding of material properties related to heat, air, moisture and to study fenestration materials and high-performance glazing technologies with emphasis on low energy materials. It also aims to provide a comprehensive understanding of sustainable and resilient construction techniques, with a focus on thermal insulation methods, moisture-resistant techniques, and fire safety measures.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand thermophysical and hygrothermal properties of building materials and analyze recent building insulation materials based on these properties	Understanding Analyzing		
CO 2	Evaluate properties of glazing materials and technologies for appropriate use	Evaluating		
CO 3	Assess the role of green building products and biomaterials in sustainable building practice	Evaluating		
CO 4	Review installation techniques related to thermal insulated, moisture resistant, fire rated and acoustic materials	Analyzing		
CO 5	Analyze case studies having green building products and innovative material in the market	Analyzing		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100 Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks End Semester Examination pattern:				

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There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN			
Module	Contents	CO	Hours
I	Introduction to building materials Classification, Properties, Applications- Building materials used in construction elements- Structural characteristics of building materials, Mechanical properties, Physical properties, Chemical properties Thermophysical properties of materials Heat Transfer Properties: Thermal Conductivity, Specific Heat, Thermal Mass, Thermal Diffusivity, Albedo Review of thermophysical properties of State-of-the-art building insulation materials, application. Nano insulation materials (NIM) and Dynamic insulation materials (DIM). Hygrothermal behaviour of materials Moisture Ingress in Building Envelope Materials, Transport Mechanisms and Practical Mitigation Approaches- Key Moisture-Related Properties of Building Materials: Moisture Absorption and Retention, Moisture Resistance, Vapor Permeability, Capillary Action, Moisture Buffering Moisture resistant building materials- vernacular and recent Epoxy moisture barrier, waterproofing materials	CO1	10
II	Fenestration Materials and Technologies Glazing Properties-Thermal conductance (U-value), Solar Heat Gain Coefficient (SHGC), Visible Light Transmittance (VT) Advanced glazing systems -Multilayer glazing, suspended films, vacuum glazing, low-emissivity coatings, smart windows (electrochromic materials), solar cell glazing, self-cleaning glazing, aerogels, glazing cavity gas fills,	CO2	9

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	glass facade systems, phase change material window products, air sandwiches, Liquid crystal windows, Thermochromic coatings, Photochromic coatings.		
III	Green building product and materials Material Passport Systems, Products with locally sourced, non-toxic, recycled low-carbon emission processes Green building products like eco block, cellulose insulation, adobe, rammed earth, bamboo-based products Insulated concrete forms (ISF), hydra form, prefabs / structural insulating panels 3D-printed sustainable materials, Metamaterials, Aerogels, smart materials, Thermally Adaptive Fabrics, Recycled steel Biomaterials from industrial, mining, agricultural waste - Use of post-consumer and industrial waste such as fly-ash, bags, building demolition waste – use of salvaged materials	CO3	9
IV	Construction techniques- Thermal insulation installation materials for walls, roofs-types, installation methods Moisture-resistant construction techniques:- Vapor barriers, Desiccants, Breathable fabrics Fire rated materials- installation Acoustic insulation- materials and installation methods Alternative construction techniques such as SMB, CSEB, and steam cured blocks, Modern sustainable Composite materials and innovative construction techniques	CO4	10
V	Building Material case studies and market study Historical, Vernacular building materials in different climates- properties Case studies using green building products and innovative building materials Material market study- latest trends, cost estimation Review of eco-friendly materials with technological innovation available in market	CO5	10

Reference Books:

1. Hugo, H. (2016). *Applied building physics: Ambient conditions, building performance and material properties*. Ernst & Sohn.
2. Shetty, M. S. (2005). *Concrete technology: Theory and practice*. S. Chand Publishing.
3. Jagadish, K. S. (2008). *Alternative building materials and technologies*. New Age International Publishers.
4. Grimvall, G. (1999). *Thermophysical properties of materials*. North-Holland Publishers.
5. Baetens, R., Jelle, B. P., & Gustavsen, A. (2010). Properties, requirements and possibilities of smart windows for dynamic daylighting and solar energy control. *Solar Energy Materials and Solar Cells*, 94(2), 87–105. Elsevier.
6. Carmody, J., Selkowitz, S., Lee, E. S., Arasteh, D., & Willmert, T. (2004). *Window systems for high-performance buildings*. W. W. Norton & Company.
7. Yudelson, J. (2008). *Green building A to Z: Understanding the language of green building*. New Society Publishers.
8. Anderson, J., Shiers, D., & Sinclair, M. (2002). *The green guide to specification* (3rd ed.). Blackwell Science.
9. Kibert, C. J. (2016). *Sustainable construction: Green building design and delivery* (4th ed.). John Wiley & Sons.
10. Abrams, D. W. (1986). *Low energy cooling: A guide to the practical application of passive cooling and cooling energy conservation measures*. Van Nostrand Reinhold Co.
11. Cornejo, D. N., & Haro, J. L. (2009). *Building materials: Properties, performance and applications*. Nova Science Publishers Inc.
12. Bureau of Energy Efficiency, Government of India. (2017). *Energy conservation building code 2017*.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251TED003	Building Physics I	3-2-0	4	2025
Course Objectives	The course serves both as an introduction to the physical processes lying behind the design of a building's envelope, interior and equipment, and as an initiation for a proper integration of technology in architecture. The course intends to provide knowledge about the prerequisites of building and the building components' function to achieve heat and light balances.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Analyze sun-earth relationship with respect to buildings	Analyzing		
CO 2	Design of shading devices	Creating		
CO 3	Evaluate the thermal performance within the built environment	Evaluating		
CO 4	Evaluate the thermal comfort of occupants within the built environment	Evaluating		
CO 5	Evaluate and design indoors for daylighting and supplementary artificial lighting	Evaluating & Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100 Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Climate and Solar Geometry Global climate, Elements of climate, Climate and its classification, Lococentric and heliocentric views, Sun path, Solar constant, Angles defining the Earth-Sun relationship, Angle of incidence, Shading design, and Passive and active shading systems. Exercises: 1. Sun path diagrams/solar charts 2. Calculation of angle of incidence 3. Design of horizontal, vertical and egg-crate shading devices	CO1, CO2	14
II	Thermal Performance of Buildings Thermal quantities, Heat transfer in buildings, Thermal behaviour of composite layers, solid envelopes, cavities, and fenestrations, Thermal bridges, Types of Insulation. Exercise/Lab: 1. Examining the thermal performance of building envelopes using suitable instruments 2. Measurements of thermal quantities 3. Heat flow calculations using Steady-state, and Periodic-state methods	CO3	14
III	Thermal Comfort Thermal balance and comfort, Factors of comfort, Adjustment mechanisms, Thermal Comfort indices and comfort zone, Thermal Comfort models, Latest developments in the field of Thermal Comfort Exercise/Lab: Thermal comfort evaluation using suitable instruments and the models listed below: 1. Heat balance model, 2. Adaptive thermal comfort model 3. Tropical Summer Index (TSI) 4. Effective Temperature (ET) 5. Standard Effective Temperature (SET)	CO4	18
IV	Daylighting Physics of Light, Photometry, Vision and Colours, Visual Comfort, Principles of lighting, Light sources, Calculation of Illuminance, Daylighting and Fenestration Design, Daylight	CO5	18

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	Autonomy (DA), Useful Daylight Illuminance (UDI), Spatial Daylight Autonomy (sDA), Annual Sunlight Exposure (ASE), Daylight Factor (DF) Exercise/Lab: 1. Examining the illumination levels of daylight in selected rooms 2. Daylighting calculation using various parameters studied 3. Daylighting design for indoors		
V	Artificial Lighting Controlling Light, Principles of controlling Light, Lighting Design concepts - Qualitative and Quantitative approaches, Lighting calculations, Presentation of spaces with light, Computer simulation of visual effects of various lamps and luminaires, Integration of artificial lighting with natural light, Permanent Supplementary Artificial Lighting of Interiors (PSALI), Energy conservation in lighting Exercise/Lab: 1. Measurements of artificial light and comparing with the standards 2. Calculation and design of artificial lighting using Permanent Supplementary Artificial Lighting of Interiors (PSALI) method 3. Market study of different types of luminaires	CO5	16

Reference Books:

1. Szokolay, S. (2014). *Introduction to architectural science: The basis of sustainable design* (3rd ed.). Routledge.
2. Olgyay, V. (2015). *Design with climate: Bioclimatic approach to architectural regionalism*. Princeton University Press.
3. Pinteric, M. (2017). *Building physics: From physical principles to international standards*. Springer International Publishing AG.
4. Lechner, N. (2015). *Heating, cooling, lighting: Sustainable design methods for architects* (4th ed.). John Wiley & Sons Inc.
5. Krishnan, A. (2017). *Climate responsive architecture: A design handbook for energy efficient buildings*. McGraw-Hill Education.
6. Hens, H. S. L. (2023). *Building physics: Heat, air and moisture: Fundamentals and engineering methods with examples and exercises* (4th ed.). Wiley - Ernst & Sohn.
7. Brown, G. Z., & DeKay, M. (2014). *Sun, wind & light: Architectural design strategies* (3rd ed.). Wiley.
8. Bureau of Indian Standards. (2016). *National building code of India 2016*.
9. Bureau of Indian Standards. (2010). *National lighting code 2010*.
10. Bureau of Energy Efficiency, Government of India. (2017). *Energy conservation building code 2017*.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251LED001	Environmental Design Lab I	0-4-0	2	2025
Course Objectives	To develop fundamental skills in climate-responsive building design through simulations, climate data analysis, and component/system-level design evaluations using software tools and lab-based experiments. It also builds foundational understanding of thermal comfort indices and indoor environmental quality, encouraging an evidence-based approach to sustainable architectural design.			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Analyze climate data to derive region-specific environmental design strategies.	Understand, Analyze		
CO 2	Simulate building envelope performance using appropriate software tools.	Apply, Evaluate		
CO 3	Evaluate daylight and ventilation strategies using digital tools.	Apply, Analyse		
CO 4	Conduct basic material and climate experiments to validate design decisions.	Apply, Evaluate		
CO 5	Integrate lab findings into the design studio project.	Apply, Create		
Mark Distribution: Continuous internal evaluation - 25 Internal Viva-Voce Examination - 15 Total marks - 40 Continuous Internal Evaluation pattern: Continuous Evaluation (Lab Exercises) - 15 marks Studio Integration Assignment - 10 marks				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Climate Data, Solar and Daylighting Analysis: • Interpretation of climatic parameters (temperature, humidity, solar radiation, wind patterns). • Use of climate analysis tools like Climate Consultant, Weather Tool etc. • Understanding psychrometric and bioclimatic charts. • Shadow analysis, sun path diagrams, and daylight availability mapping. Hands on Exercises: • Climate data analysis using climate charts (sun path, psychrometric, wind rose). • Generation and interpretation of climate-responsive design strategies using software • Tools such as SketchUp + Heliodon, Ladybug for Grasshopper, and Velux Daylight Visualizer.	CO1	12
II	Building Envelope Performance Analysis: Hands-on experiments and digital simulations to evaluate: • Wall-roof insulation • U-values and thermal mass • Glazing and shading device effectiveness Hands-on Exercises: • Use of current tools like THERM, Design-Builder, and Envi-met. • Envelope simulations for Studio-I design projects	CO2	12
III	Thermal Comfort & Ventilation Modelling • Introduction to adaptive comfort models and standards (ASHRAE 55, EN 15251). • Simulating natural ventilation and airflow using tools like CFD, DesignBuilder or any other available software tools. • Linking thermal comfort data with envelope design for optimization. Hands-on Exercises Designing and testing daylight and ventilation strategies for studio projects	CO3	12
IV	Material and Component Testing	CO4	12

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	• Introduction to thermal conductivity meters, emissivity measurement, and surface temperature checks using infrared thermography. • Study of local and alternative materials suitable for climate-responsive design. Hands-on Exercises • Lab experiments for material thermal properties (R-value testing, emissivity, albedo) • Fieldwork: Use of data loggers, thermometers, and anemometers for site microclimate recording • Comparative analysis between simulation predictions and field data		
V	Integration with Studio Projects: • Application of lab outputs to support envelope, façade, roof, and component-level design proposals. • Critical evaluation of passive design strategies using evidence-based analysis.	CO5	16

Reference Books:

1. Watson, D., & Labs, K. (1983). *Climatic building design: Energy-efficient building principles and practices*. McGraw-Hill.
2. Lechner, N. (2015). *Heating, cooling, lighting: Sustainable design methods for architects* (4th ed.). John Wiley & Sons.
3. Clarke, J. A. (2007). *Energy simulation in building design*. Routledge.
4. Augenbroe, G. (2004). *Advanced building simulation* (1st ed.). Routledge.
5. Garg, V., Mathur, J., Tetali, S., & Bhatia, A. (2017). *Building energy simulation: A workbook using DesignBuilder™* (1st ed.). CRC Press.
6. Brackney, L., Parker, A., Macumber, D., & Benne, K. (2018). *Building energy modelling with OpenStudio: A practical guide for students and professionals*. Springer International Publishing.
7. Ruck, N. C. (1989). *Building design and human performance*. Van Nostrand Reinhold.
8. Singh, R., Ravache, B., & Sartor, D. (2018). *Building innovation: A guide for high-performance energy efficient buildings in India*. Lawrence Berkeley National Laboratory (LBNL).
9. Krauthaim, M. (2014). *City and wind: Climate as an architectural instrument*. DOM Publishers.
10. Bureau of Energy Efficiency. (2009). *Energy Conservation Building Code (ECBC) user guide*. New Delhi: Bureau of Energy Efficiency.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
251SED001	Environmental Design Studio I	0-0-9	9	2025
Course Objectives	The aim is to introduce the students to climate and surroundings as an important aspect of environmental design - To understand in depth, the environmental factors affecting human comfort and creating comfort conditions along with the associated building physics. - To develop the ability to design built environments that are responsive to climate and microclimatic factors of the context through informed decisions at component, envelope, and building system levels.			
Syllabus	This studio introduces students to the fundamentals of climate-responsive architectural design through an environmentally conscious lens. Emphasis is placed on understanding the climatic context of a site and translating it into spatial, formal, and technical design strategies. Students engage in the analysis of climate data, site response, and passive design principles as the foundation for building design. The studio progresses from macro-level considerations like site planning and orientation to micro-level design of building components—such as facades, roofs, openings, and shading devices—that respond to the local climate. Through iterative studio exercises, students will explore: - Site analysis using climatic data (sun path, wind, humidity, rainfall) - Form finding based on passive strategies - Detailed envelope and component design - Material selection with thermal and environmental performance criteria - Use of climate simulation tools (for solar, daylight, and airflow studies)			
Course Outcomes: After the completion of the course, the students shall be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Interpret climate data and site-specific environmental conditions to inform the conceptual development of climate-responsive design.	Understand		
CO 2	Apply principles of passive design such as solar orientation, shading, natural ventilation, and daylighting in the development of architectural form and massing.	Apply		

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CO 3	Design building envelope and facade components with appropriate material and construction systems to enhance thermal comfort and environmental performance.	Create
CO 4	Analyse the impact of roof, facade, and component design decisions using environmental simulation tools (thermal, daylight, airflow) to optimize performance.	Analyze
CO 5	Integrate climate-sensitive strategies in schematic designs through iterative processes that balance aesthetics, performance, and context.	Create
Mark Distribution: Continuous internal evaluation - 200 Final Jury Evaluation (Viva) - 100 Total Marks - 300		

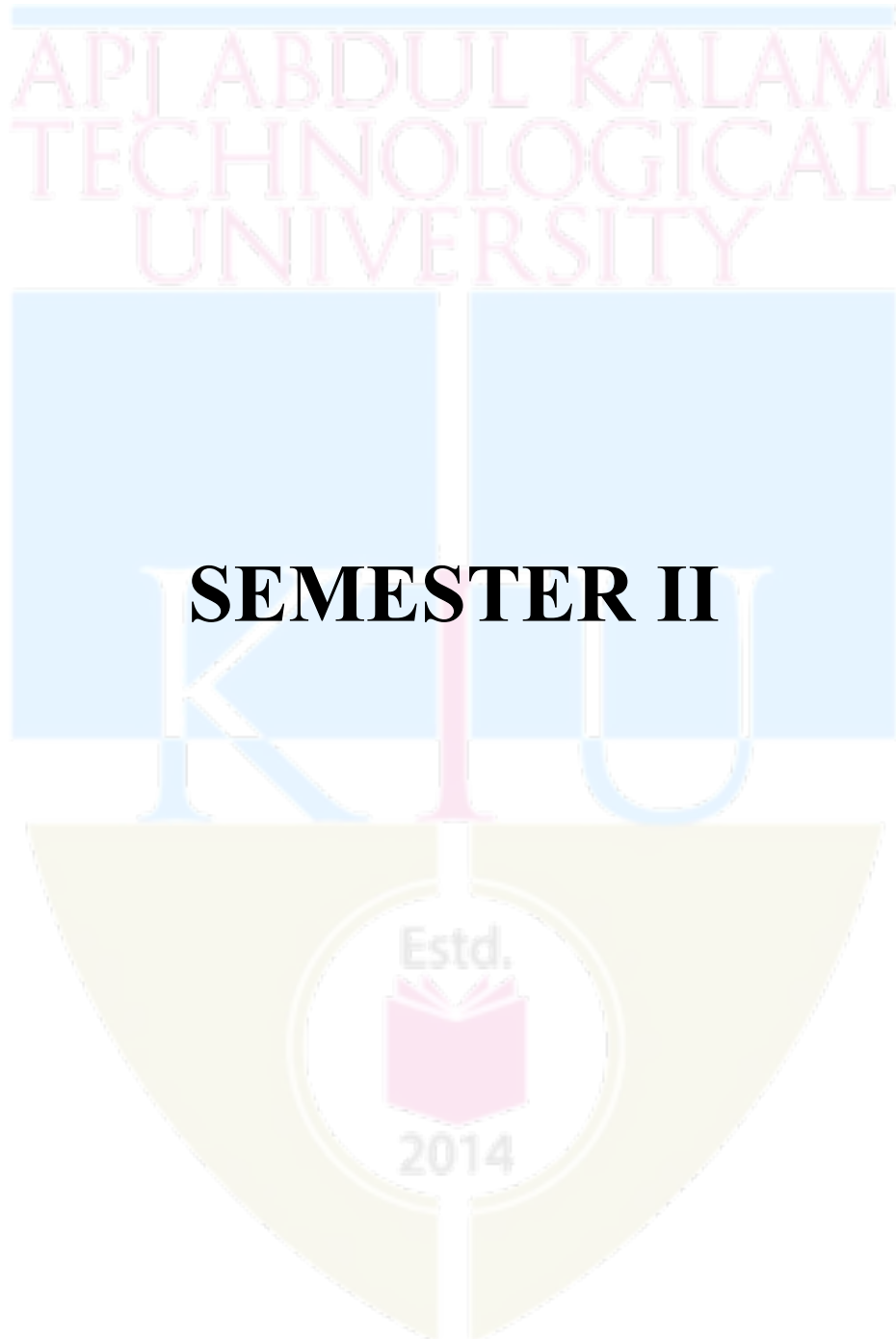
COURSE PLAN			
Module	Contents	CO	Hours
I	Understanding Climate and Context Site climate analysis, microclimatic mapping, contextual response - Identify key climate parameters and their impact on built form. - Prepare site-level microclimate assessments (sun path, wind, shading, humidity). Exercises: - Climate data analysis using climate charts (sun path, psychrometric, wind rose). - Site visits and microclimate documentation. - Study of vernacular climate-responsive strategies relevant to site climate.	CO1	24
II	Component-Level Design for Comfort Design of openings, walls, shading devices, and passive systems - Explore and design building components for climate moderation. - Understand thermal, daylight, and ventilation performance of components. Exercises:	CO2	24

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	• Detailing of climate-adaptive window and shading systems. • Wall assembly design for insulation, thermal mass, and moisture control. • Comparative analysis of traditional and innovative façade components. • 1:1 mock-up or scaled model of a responsive building component.		
III	Envelope & Façade System Design Passive design of building envelope and façade strategies • Design envelopes that support comfort, energy efficiency, and aesthetics. • Integrate passive cooling/heating strategies (e.g., cavity walls, ventilated façades). Exercises: • Envelope performance simulation using relevant software. • Propose an envelope solution for a selected building typology (institutional, residential, etc.). • Facade articulation integrating materiality, shading, and ventilation strategies.	CO3	42
IV	Climate-Responsive Building Design Project Comprehensive design of a medium-scale built form with a strong emphasis on contextual responsiveness, including rural and semi-urban settings. The project should demonstrate the integration of component, envelope, and system design strategies into a coherent, climate-responsive architectural proposal that reflects both environmental and socio-cultural sensitivities of the site. Deliverables: • Site planning with microclimate modulation (e.g., landscape features, orientation, transitional spaces, natural buffers), especially in rural topographies and settlement patterns, • Building form and layout optimizing passive strategies suited to local climatic and cultural practices. • Detailed envelope and roof system design with climate-specific material selection. • Integration of vernacular knowledge and resource-efficient technologies in the design. • Final presentation with physical / digital models, environmental simulations (e.g., daylighting, ventilation), working drawings, and a brief rural context analysis.	CO4 CO5	54

Reference Books:

1. Szokolay, S. V. (2008). *Introduction to architectural science: The basis of sustainable design* (2nd ed.). Architectural Press.
2. Hyde, R., Watson, S., Cheshire, D., & Thomson, M. (2007). *The environmental brief: Pathways for green design*. Taylor & Francis.
3. Brown, G. Z., & DeKay, M. (2014). *Sun, wind & light: Architectural design strategies* (3rd ed.). Wiley.
4. Yannas, S. (1994). *Solar energy and housing design: Principles, technology and examples*. Architectural Association.
5. Baker, N., & Steemers, K. (2000). *Energy and environment in architecture: A technical design guide*. Taylor & Francis.
6. Watson, D., & Labs, K. (1983). *Climatic building design: Energy-efficient building principles and practices*. McGraw-Hill.
7. Lechner, N. (2015). *Heating, cooling, lighting: Sustainable design methods for architects* (4th ed.). John Wiley & Sons.
8. Clarke, J. A. (2007). *Energy simulation in building design*. Routledge.
9. Augenbroe, G. (2004). *Advanced building simulation* (1st ed.). Routledge.
10. Garg, V., Mathur, J., Tetali, S., & Bhatia, A. (2017). *Building energy simulation: A workbook using DesignBuilder™* (1st ed.). CRC Press.
11. Brackney, L., Parker, A., Macumber, D., & Benne, K. (2018). *Building energy modelling with OpenStudio: A practical guide for students and professionals*. Springer.
12. Ruck, N. C. (1989). *Building design and human performance*. Van Nostrand Reinhold.
13. Singh, R., Ravache, B., & Sartor, D. (2018). *Building innovation: A guide for high-performance energy efficient buildings in India*. Lawrence Berkeley National Laboratory (LBNL).
14. Krauthaim, M. (2014). *City and wind: Climate as an architectural instrument*. DOM Publishers.
15. Bureau of Energy Efficiency. (2009). *Energy Conservation Building Code (ECBC) user guide*. Bureau of Energy Efficiency, Government of India.



SEMESTER II

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252TED001	Renewable Energy and Building Systems	2-1-0	3	2025
Course Objectives	The course intends to create a deep understanding on the principles of mechanical services, the recent trends within the context of energy efficiency and energy performance. It aims to create understanding of different renewable energy technologies, principles and applicability as well as its integration with building management systems so as to create net zero design.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Calculate cooling and heat load and design duct sizes	Analysing Creating		
CO 2	Analyse and design in fire services, plumbing, sanitation and vertical transportation in built environment	Analysing Creating		
CO 3	Assess role of Integrated building management system and the need	Understanding		
CO 4	Examine and review various renewable energy applications	Analysing		
CO 5	Review case studies of renewable energy systems in completed projects and net zero buildings	Analysing		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Mechanical Services: HVAC HVAC: - Psychrometrics, Basics of HVAC Seasonal Energy Efficiency Ratio (SEER). Heat Recovery Ventilation (HRV) and Energy Recovery Ventilation (ERV), VAV and VRF systems Recent trends in air-conditioning- Smart HVAC, DeVAP, Ductless air conditioning Systems, Motion activated, thermally driven air conditioning, AI-powered systems, Absorption Refrigeration System, advanced and hybrid heat pump systems, Radiant HVAC System, Dual duct system, Chilled beams and ceilings, low/hybrid energy cooling systems, Displacement and Underfloor Air Distribution (UFAD) Eco-friendly refrigerants and advanced filtration technologies, dual inverter systems Cooling and Heat Load Calculation using ASHRAE Standards Duct Design- Calculation of duct sizes, ISHRAE standards	CO1	10
II	Mechanical Services: Fire services, plumbing, sanitation and vertical transportation systems Fire services: General requirements of fire resisting buildings as per IS and NBC, Fire protection requirement for high rise buildings. Effect of fire load & ventilation condition Plumbing and Sanitation: Distribution Systems in Multi-Storied Buildings, Pipe sizing by formula, Pipe sizes and resistances, Innovative sanitation technologies, Sewage collection systems and recycling of water, treatment and disposal of sewage, Integrated solid waste handling system- Solid waste disposal Vertical transportation Elevators- Traffic Analysis and design- Handling capacity, Planning and design of elevator lobby areas, New generation	CO2	10

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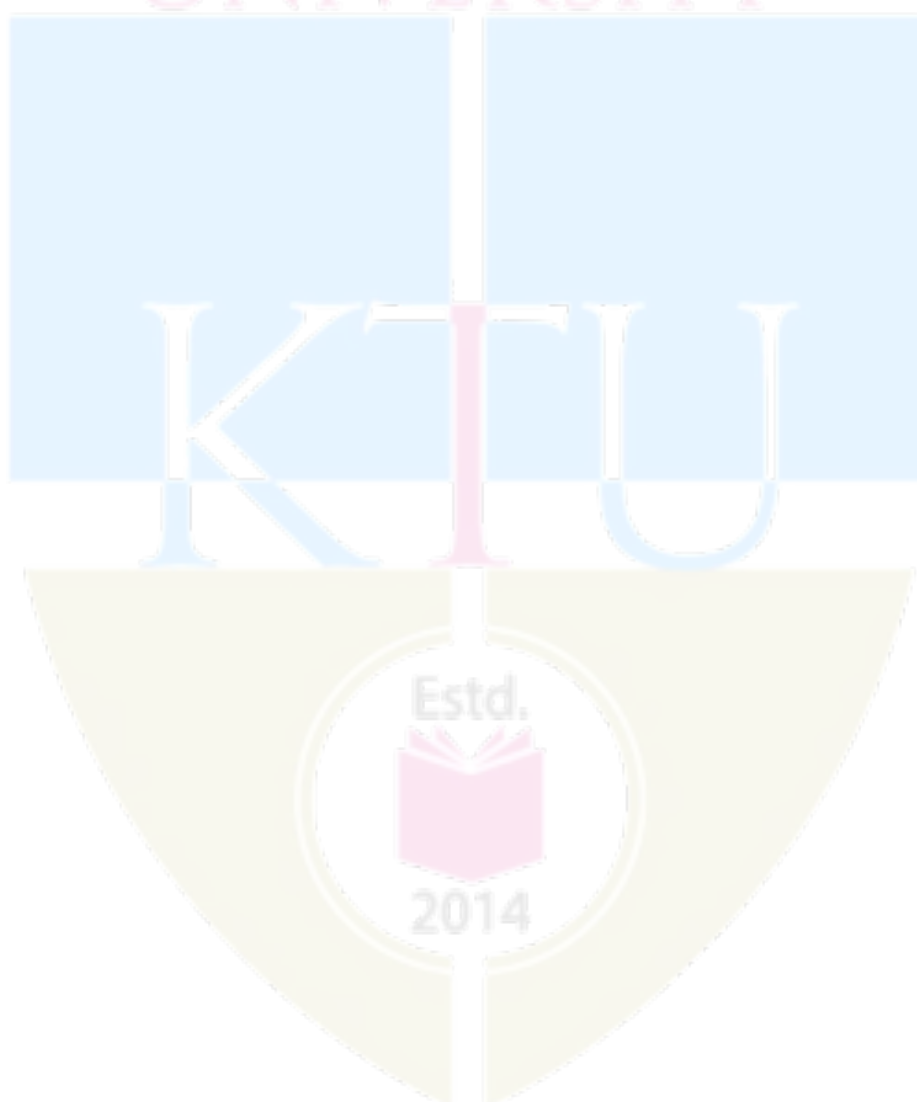
	vertical transportation concepts, recent developments on elevator and escalator technology		
III	Energy management systems Environmental Impact of HVAC Systems, Need for Integration of services, Computerized HVAC Systems - Direct Digital Control, BTU monitoring & control, Data networking– IBMS system and its components	CO3	8
IV	Renewable Energy systems Renewable energy sources, Introduction to renewable energy systems, need for net zero design, design for climate, Solar energy: Computation of solar radiation on horizontal and inclined surfaces, Solar Cooling and heating, Solar thermal power generation, Solar absorption air-conditioning, Solar powered devices, Solar HVAC Panel, Building integrated photovoltaic systems. Geothermal HVAC systems, Use of green hydrogen and fuel cells. Advanced energy storage systems. Application of phase change materials (PCM).	CO4	10
V	Integration of renewable energy systems in building design Introduction to smart grid and Artificial Intelligence (AI) systems. Case studies of renewable energy systems in completed projects and net zero buildings. Field trips to review installations, Performance assessment of installed systems, Determination of on-site renewable energy generation potential, Review of integration issues for renewable technologies.	CO5	10

Reference Books:

1. Haines, R. W., & Myers, M. E. (2010). *HVAC systems design handbook* (5th ed.). McGraw-Hill Education.
2. American Society of Heating, Refrigerating and Air-Conditioning Engineers. (2024). *2024 ASHRAE handbook: HVAC systems and equipment* (SI ed.). ASHRAE.
3. Angel, W. L. (2020). *HVAC design sourcebook* (2nd ed.). McGraw Hill.
4. Gupta, N. C. (2016). *Comprehensive HVAC design: A handbook on practical approach to air conditioning, heating and ventilation system*. Viva Books.
5. Bureau of Indian Standards. (2016). *National building code of India 2016: Part 4 fire & life safety* (Vols. 1–2). Bureau of Indian Standards.

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6. Peter, Y. (2024). *Renewable energy: Challenges and solutions*. Springer International Publishing.
7. Sukhatme, S. P., & Nayak, J. K. (2017). *Solar energy: Principles of thermal collection and storage* (4th ed.). McGraw Hill Education.
8. DeKay, M., & Brown, G. Z. (2014). *Sun, wind, and light: Architectural design strategies* (3rd ed.). John Wiley & Sons.
9. Hodge, B. K. (2017). *Alternative energy systems and applications* (2nd ed.). John Wiley & Sons.
10. Kishore, V. V. N. (2008). *Renewable energy engineering and technology: Principles and practice*. Routledge.
11. Arthur, A., et al. (2016). *HVAC equations, data, and rules of thumb* (3rd ed.). McGraw-Hill Education.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252TED002	Building Physics II	2-1-0	3	2025
Course Objectives	The course is intended to provide knowledge about the scientific principles behind moisture, ventilation and acoustics in buildings. It integrates science and technology into architecture for the design of a building's envelope, interior and equipment. It also aims to develop the ability to apply them to analyze key design processes.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome			Learning level
CO 1	Design of building components for and against moisture			Creating
CO 2	Evaluating the built environment to optimize natural, mechanical and hybrid ventilation systems of ventilation			Analyzing and evaluating
CO 3	Examining the acoustic comfort within the built environment			Evaluating
CO 4	Design of building components or spaces for enhanced acoustic performance			Creating
CO 5	Review of case studies on sustainable concepts and environmental design firms			Understanding
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.				

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Module	Contents	CO	Hours
I	Moisture in buildings Sources of moisture, Moisture transport mechanisms, Moisture damage in buildings, Health implications of dampness, Moisture tolerant materials and systems, Humidity control design principles	CO1	9
II	Planning for Ventilation Functions of ventilation, Ventilation requirement calculations – orientation, position of openings, size of openings, controls of openings, Infiltration, Requirements of ventilation as per ASHRAE/ECBC, Artificial ventilation techniques – forced ventilation, fresh air systems, demand-controlled ventilation etc.	CO2	9
III	Building Acoustics Room Acoustics - Behaviour of sound inside a room, Acoustic defects, Ripple tank studies, Room modes, Acoustic reflectors, Shape of hall, Seating Arrangements, Volume, Reverberation time measurements, Absorption coefficient measurements, Types of absorbents, Assisted Resonance, Measurement of sound power levels - anechoic and reverberant rooms, Design strategies for classroom, Auditorium and amphitheatre acoustics	CO3, CO4	10
IV	Noise and Vibration Noise level, Sound Paths, Sound Transmission - Airborne and Structure borne Sound, Air borne sound insulation, Structure borne sound insulation, Vibration, Types of vibration, Measurement of vibration, Transmission Loss, Mass Law (Berger's Law), Frequency Law, Noise control, Environmental Noise – Anti-noise barriers, Urban acoustics	CO3, CO4	10
V	Sustainable Building Concepts and Case studies Environmental design & cultural identity – Contemporary examples from tropical countries, building concepts aligned with Sustainable Development Goals, Study and analysis of implemented sustainable ideas such as Earthships, Passivhaus, etc., Study of emerging sustainability concepts such as Net Zero Carbon buildings, Carbon-Neutral buildings, Carbon positive buildings, etc., Review of a national or international level firm practicing Environmental Design	CO5	10

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Reference Books

1. Szokolay, S. (2014). *Introduction to architectural science: The basis of sustainable design* (3rd ed.). Routledge.
2. Hens, H. S. L. (2023). *Building physics – Heat, air and moisture: Fundamentals and engineering methods with examples and exercises* (4th ed.). Wiley - Ernst & Sohn.
3. Egan, M. D. (2009). *Architectural acoustics* (1st ed.). Cengage Learning.
4. Jain, V. K. (2022). *Architectural acoustics* (1st ed.). Laxmi Publications Pvt. Ltd.
5. Pinteric, M. (2017). *Building physics: From physical principles to international standards*. Springer International Publishing AG.
6. Baker, N., & Steemers, K. (2000). *Energy and environment in architecture: A technical design guide*. E & FN Spon.
7. Ermann, M. (2015). *Architectural acoustics illustrated*. Wiley.
8. Brown, G. Z., & DeKay, M. (2014). *Sun, wind & light: Architectural design strategies* (3rd ed.). Wiley.
9. Krishnan, A. (2017). *Climate responsive architecture: A design handbook for energy efficient buildings*. McGraw-Hill Education.
10. Bureau of Indian Standards. (2016). *National building code of India 2016*.
11. Bureau of Energy Efficiency, Government of India. (2017). *Energy conservation building code 2017*.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252RED001	Research Methodology and Seminar	2-1-0	2	2025
Course Objectives	To introduce students to the foundations and paradigms of research in environmental design. To introduce students to research approaches relevant to sustainable and resilient environments. To enable formulating research problems in environmental performance, planning, and socio-spatial dynamics. To develop competency in analytical tools and techniques, including statistical and spatial analysis. To apply research tools and ethical practices for scholarly and professional dissemination			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand the basics of research and identify and formulate research problems in environmental design	Understanding & Analyzing		
CO 2	Analyze existing literature to identify research gaps within environmental design contexts.	Analyzing		
CO 3	Design a research framework responsive to ecological, social, and spatial parameters.	Creating		
CO 4	Apply appropriate data collection and analysis techniques in environmental design research	Applying		
CO 5	Utilize statistical software and GIS tools to analyze environmental and spatial data.	Applying		
CO 6	Produce high-quality academic outputs such as theses and scholarly papers that reflect research findings.	Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				

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Continuous Internal Evaluation pattern:

Assignment/ Quiz/ Course project - 25 marks

Internal test - 15 marks

End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

COURSE PLAN

Module	Contents	CO	Hours
I	Foundations of Research in Environmental Design • Introduction to research: Definitions, objectives, types • Interdisciplinary nature of environmental design research- Sustainability, resilience, and human-environment interactions as research themes • Paradigms: Positivist, Interpretivist, Pragmatic, ecological determinism. • Structure of a research study: Research questions, hypotheses, and aim and objectives	CO1	10
II	• Need for Statistical methods in research, Sampling methods and sample size determination, Basic statistics for environmental data (descriptive, inferential, correlational) Introduction to SPSS Descriptive Statistics (Foundational) (Using software) • Measures of central tendency: Mean, Median, Mode • Measures of dispersion: Standard Deviation, Variance, Range • Frequency distribution and cross-tabulation • Histograms and bar charts (for perception survey data, usage patterns, etc.) Inferential Statistics (For Hypothesis Testing) • T-tests (compare two groups: e.g., daylight performance in two building types) • Chi-square tests (categorical data: e.g., material use vs. function)	CO1 CO4 CO5	9

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	ANOVA (Analysis of Variance) (multiple group comparisons: e.g., thermal comfort across zones) (To be handled by faculty from the Mathematics department)		
III	Literature Review and Theoretical Frameworks - Role and techniques of literature review - Mapping knowledge domains, identifying gaps and framing research questions - Conceptual and theoretical framework development Research Design and Methodologies - Quantitative, qualitative and mixed methods with environmental case applications- Case study, empirical, analytical, simulation-based, action research	CO2 CO3	10
IV	Data Collection Techniques - Climate data, land-use data, behavioural observations, energy and resource audits - Surveys, interviews, focus groups, observations - Design of questionnaires and interview guides Research Reporting and Ethics - Structure and components of research reports - Writing styles and visual representation of data - Citation styles and reference management tools- Use of reference management software like Mendeley, Zotero, EndNote, etc - Research ethics, plagiarism, and IPR - Preparation of thesis, conference paper, and journal article	CO4 CO6	9
V	Data Analysis - Correlation and Regression Analysis: - Pearson/Spearman correlation - Linear regression (e.g., predicting energy use based on building form) and multiple regression (to include multiple variables such as orientation, material, occupancy), Analytic Hierarchy Process (Statistical analysis using the Software SPSS) (To be handled by faculty from the Mathematics department)	CO5 CO6	10

Reference Books

1. Creswell, J. W. (2014). *Research design: Qualitative, quantitative, and mixed methods approach* (4th ed.). SAGE Publications.
2. Denzin, N. K., & Lincoln, Y. S. (Eds.). (2018). *The SAGE handbook of qualitative research* (5th ed.). SAGE Publications.
3. Groat, L., & Wang, D. (2013). *Architectural research methods* (2nd ed.). Wiley.
4. Hardle, W. K., Simar, L., & Fengler, M. R. (2017). *Applied multivariate statistical analysis* (4th ed.). Springer.
5. Kothari, C. R. (2004). *Research methodology: Methods and techniques* (2nd ed.). New Age International Publishers.
6. Leedy, P. D., & Ormrod, J. E. (2019). *Practical research: Planning and design* (12th ed.). Pearson.
7. Miles, M. B., Huberman, A. M., & Saldaña, J. (2014). *Qualitative data analysis: A methods sourcebook* (3rd ed.). SAGE Publications.
8. Tabachnick, B. G., & Fidell, L. S. (2019). *Using multivariate statistics* (7th ed.). Pearson.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252LED001	Environmental Design Lab II	0-4-0	2	2025
Course Objectives	• To equip students with advanced tools and techniques to assess energy performance in buildings. • To enable life cycle analysis of materials and systems in the context of Net Zero Energy Buildings (NZEB). • To introduce the students to energy compliance rating systems • To provide simulation support for studio-based design decisions related to energy efficiency, renewable systems, and Low Carbon buildings.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Interpret building energy simulation outputs and analyze energy loads using industry-standard tools.	Apply		
CO 2	Evaluate passive and hybrid strategies for climate-responsive and energy-efficient design	Analyze		
CO 3	Assess the environmental impact of building materials using LCA methodologies and software.	Evaluate		
CO 4	Interpret and apply national and international energy rating system requirements for sustainable design certification.	Create		
CO 5	Integrate lab analyses with studio design projects to propose environmentally responsive and certifiable design solutions.	Evaluate		
Mark Distribution: Continuous internal evaluation - 25 Internal Viva-Voce Examination - 15 Total marks - 40 Continuous Internal Evaluation pattern: Continuous Evaluation (Lab Exercises) - 15 marks Studio Integration Assignment - 10 marks				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Energy Modelling and Load Analysis • Introduction to building energy modelling concepts • Internal and external load calculations • Climate-based HVAC and lighting loads • Passive and Active System Integration Exercises: • Hands-on sessions using EnergyPlus/ DesignBuilder/ OpenStudio or any currently available software tools. • Modelling Passive, Active and Hybrid systems and evaluating their effectiveness	CO1	10
II	Life Cycle Analysis (LCA) • Overview of LCA in environmental building design • Material selection based on embodied energy and carbon footprint • Introduction to tools such as One Click LCA / Sima pro/Athena or any currently available tools. Exercises: • Lifecycle impact comparison using databases (e.g., eco invent, OneClick LCA). • Detail exploration of low-carbon construction assemblies. • Documentation of material choices with justification from LCA.	CO2	10
III	Building Energy Rating Systems and Certification Tools • Introduction to national and international green building rating systems: GRIHA (India), LEED (US), BREEAM (UK), WELL, EDGE, IGBC • Comparative framework of rating systems: structure, point systems, certification pathways • Core focus areas: energy efficiency, water management, material usage, IEQ, innovation • Understanding compliance tools and documentation standards • Introduction to GRIHA and LEED energy simulation requirements Exercises: • Simulation-based credit assessment using LEED and GRIHA templates • Energy performance calculations for case projects aligned to rating frameworks	CO3	12

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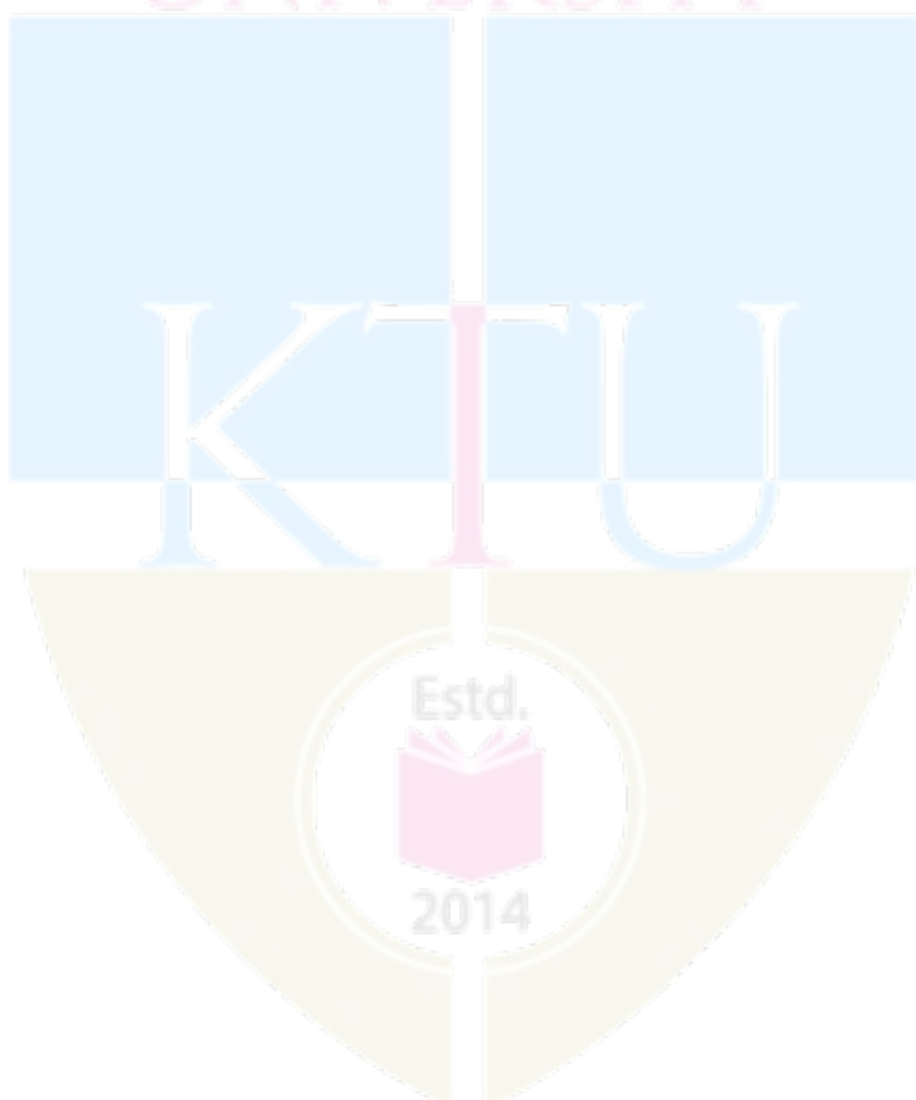
	Preparation of rating documentation and compliance checklist		
IV	Net Zero Performance Assessment and Compliance - Simulation of NZEBs for site-specific climatic conditions - Developing energy balance sheets (generation vs. consumption) - Optimization of envelope and system parameters for NZEB targets Exercises: - Net Zero performance dashboard (renewables vs demand, carbon vs sequestration).	CO4	12
V	Integration with Studio Projects - Mapping lab-based simulation results to design decisions - Evaluating the impact of proposed interventions on energy and carbon targets - Portfolio preparation and technical documentation support the studio.	CO5	20

Reference Books

1. Clarke, J. A. (2007). *Energy simulation in building design*. Routledge.
2. Augenbroe, G. (2004). *Advanced building simulation* (1st ed.). Routledge.
3. Garg, V., Mathur, J., Tetali, S., & Bhatia, A. (2017). *Building energy simulation: A workbook using DesignBuilder™* (1st ed.). CRC Press, Taylor & Francis Group.
4. Brackney, L., Parker, A., Macumber, D., & Benne, K. (2018). *Building energy modelling with OpenStudio: A practical guide for students and professionals*. Springer International Publishing.
5. Pacheco-Torgal, F., Mistretta, M., Kaklauskas, A., Granqvist, C. G., & Cabeza, L. F. (Eds.). (2013). *Nearly zero energy building refurbishment: A multidisciplinary approach*. Springer International Publishing AG.
6. Singh, R., Ravache, B., & Sartor, D. (2018). *Building innovation: A guide for high-performance energy efficient buildings in India*. Lawrence Berkeley National Laboratory (LBNL).
7. Bureau of Energy Efficiency. (2009). *Energy Conservation Building Code (ECBC) user guide*. New Delhi: Bureau of Energy Efficiency.
8. Keeping, M., & Shiers, D. (2017). *Sustainable building design: Principles and practice*. John Wiley & Sons.
9. Kwok, A. G., & Grondzik, W. T. (2018). *The green studio handbook: Environmental strategies for schematic design* (3rd ed.). Routledge.

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10. U.S. Green Building Council (USGBC). (2016). *Whole-building life cycle assessment: Reference and guidance*. USGBC.
11. The Energy and Resources Institute (TERI). (2010). *GRIHA manuals* (Vols. 1–5). TERI Press.
12. ASHRAE. (2023). *ASHRAE handbook – HVAC applications*. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc.
13. Curran, M. A. (Ed.). (2012). *Life cycle assessment handbook: A guide for environmentally sustainable products*. Scrivener Publishing/Wiley.
14. U.S. Green Building Council (USGBC). (2013). *LEED v4 for building design and construction: Reference guide*. USGBC.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252SED001	Environmental Design Studio II	0-0-8	8	2025
Course Objectives	To enable students to design buildings and environments that approach or achieve Net Zero Energy and Carbon targets by integrating site-responsive planning, low-carbon materials, passive-active energy strategies, carbon sequestration potentials, and life cycle assessments. Emphasis will be on a performance-driven design approach combining systems thinking and ecological intelligence.			
Syllabus	This studio builds on the foundations of climate-responsive design by taking students into the domain of net zero energy and low-carbon architecture. It introduces an integrated approach to energy-conscious design by examining buildings through the lens of carbon footprint, embodied energy, and life cycle assessment (LCA). The studio challenges students to develop building design proposals that strategically balance operational energy and embodied energy to move toward net zero targets. This includes analysing site-specific renewable energy potential, vegetation and carbon sequestration, and ecological footprints. Key Focus Areas: • Net zero energy principles in site and building design • Life cycle analysis of materials, components, and systems • Carbon sequestration strategies through landscape and design interventions • Integration of active and passive systems for energy optimization • Simulation-based evaluation of design performance Through real or hypothetical site contexts, students explore ways to integrate architecture with systems thinking, examining the interplay of form, technology, materials, and natural processes in pursuit of sustainable, performance-driven design.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Analyze the energy and carbon context of a site including renewable potential, microclimate, vegetation, and ecological systems.	Analyze		
CO 2	Design building envelopes and components that integrate passive and active energy strategies to reduce operational energy demand.	Create		

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CO 3	Apply principles of life cycle assessment (LCA) to evaluate and justify material and system choices based on embodied energy and carbon footprint.	Apply
CO 4	Develop integrated building design proposals that target net zero energy and carbon balance, supported by simulation tools and data-driven design.	Create
CO 5	Evaluate climate-responsive and environmentally optimized design solutions through technical documentation, models, and performance metrics.	Evaluate
Mark Distribution: Continuous internal evaluation - 200 Final Jury Evaluation (Viva) - 100 Total Marks - 300		

COURSE PLAN			
Module	Contents	CO	Hours
I	Energy and Carbon in the Built Environment: Site-energy context, carbon mapping, ecological design base • Understand energy use and carbon emissions across building lifecycle stages. • Identify carbon sequestration potential through vegetation and landform. • Evaluate site-level energy flows and renewable potential (solar, wind, biomass). Exercises: • Site energy and carbon mapping exercises. • Analysis of ecosystem services and landscape strategies. • Case studies of Net Zero and Net Positive campus/building projects.	CO1	14
II	Low-Carbon and Energy-Efficient Building Systems: Building envelope, HVAC integration, renewable systems, passive-active hybrids • Design low-energy envelope systems integrating passive cooling/heating.	CO2	14

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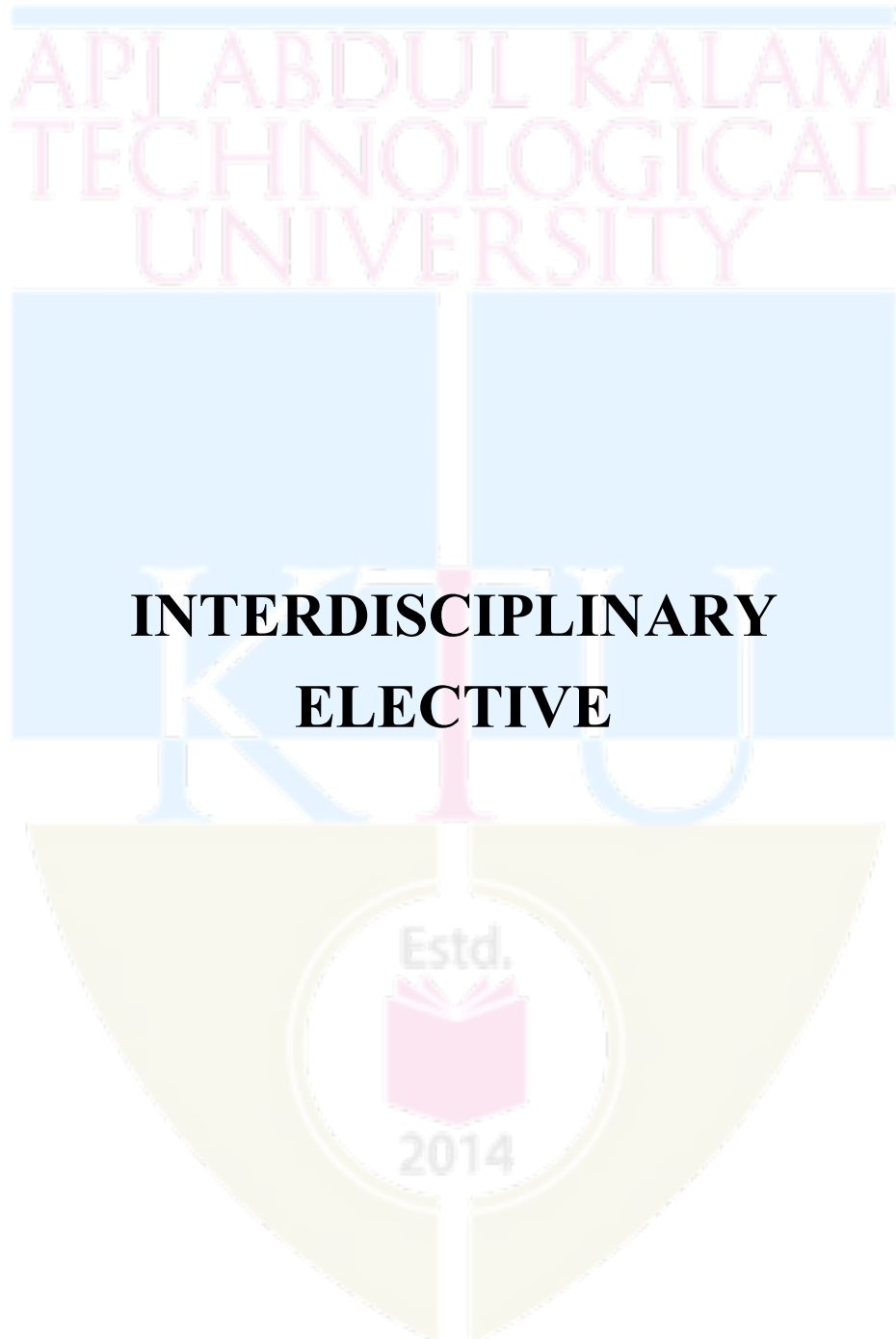
	- Explore optimal sizing and integration of renewable energy technologies. - Develop strategies for natural ventilation, daylighting, and efficient HVAC systems. Exercises: - Simulation exercises using relevant energy modelling software. - Envelope detailing for thermal performance and energy efficiency. - Energy performance benchmarking (EPI, EUIs, etc.).		
III	Life Cycle Thinking and Material Intelligence: Embodied energy, LCA of materials and systems, circularity - Apply LCA tools to compare materials based on energy and carbon footprint. - Design with resource-efficiency and end-of-life reuse or recyclability in mind. - Evaluate operational vs embodied impacts to inform system design decisions. Exercises: - Lifecycle impact comparison using databases such as eco invent, OneClick LCA etc. - Detail exploration of low-carbon construction assemblies. - Documentation of material choices with justification from LCA.	CO3	38
IV	Integrated Net Zero Design Studio Project: Holistic design of a Net Zero Energy Building (or Near Zero), e.g., Innovation Hub, Research Centre, or Institutional Facility. - Integrate all modules into a comprehensive design that meets Net Zero goals. - Demonstrate performance-based decision making across scales - from site to systems. - Effectively communicate energy/carbon metrics and ecological strategies. Exercises: - Complete architectural design including site planning, envelope, systems. - Net Zero performance dashboard (renewables vs demand, carbon vs sequestration).	CO4, CO5	62

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	• Technical documentation, environmental simulations, and physical/digital models.		
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Reference Books

1. Baker, N., & Steemers, K. (2000). *Energy and environment in architecture: A technical design guide*. Taylor & Francis.
2. Watson, D., & Labs, K. (1983). *Climatic building design: Energy-efficient building principles and practices*. McGraw-Hill.
3. Lechner, N. (2015). *Heating, cooling, lighting: Sustainable design methods for architects* (4th ed.). John Wiley & Sons.
4. Ruck, N. C. (1989). *Building design and human performance*. Van Nostrand Reinhold.
5. Singh, R., Ravache, B., & Sartor, D. (2018). *Building innovation: A guide for high-performance energy efficient buildings in India*. Lawrence Berkeley National Laboratory (LBNL).
6. Krautheim, M. (2014). *City and wind: Climate as an architectural instrument*. DOM Publishers.
7. Bureau of Energy Efficiency. (2009). *Energy Conservation Building Code (ECBC) user guide*. New Delhi: Bureau of Energy Efficiency.
8. Torgal, F. P., Mistretta, M., Kaklauskas, A., Granqvist, C. G., & Cabeza, L. F. (Eds.). (2013). *Nearly zero energy building refurbishment: A multidisciplinary approach*. Springer.
9. Keeping, M., & Shiers, D. (2017). *Sustainable building design: Principles and practice*. John Wiley & Sons.
10. Kwok, A. G., & Grondzik, W. T. (2018). *The green studio handbook: Environmental strategies for schematic design* (3rd ed.). Routledge.
11. U.S. Green Building Council. (2016). *Whole-building life cycle assessment: Reference and guidance*. USGBC.
12. The Energy and Resources Institute (TERI). (2010). *GRIHA manuals* (Vols. 1–5). TERI Press.
13. ASHRAE. (2023). *ASHRAE handbook: HVAC applications*. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc.
14. Curran, M. A. (Ed.). (2012). *Life cycle assessment handbook: A guide for environmentally sustainable products*. Scrivener Publishing/Wiley.
15. U.S. Green Building Council. (2013). *LEED v4 for building design and construction: Reference guide*. USGBC.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EED001	Green Building Rating Systems	2-1-0	3	2025
Course Objectives	The course intends to create an understanding of the principles and importance of green buildings and to familiarize with various green building rating systems used globally and in India. It also aims to equip the students with the skill to evaluate building performance based on green building criteria and carry out assessments by applying rating system criteria to case studies or real projects.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand the principles of sustainable planning and design, and the relevance of green buildings	Understanding		
CO 2	Analyze various green building rating systems used globally and in India	Analyzing		
CO 3	Evaluate building performance based on sustainability metrics	Evaluating		
CO 4	Apply rating system criteria to case studies or real projects	Applying		
CO 5	Derive and devise strategies that will improve the sustainability of a given building	Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.				

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COURSE PLAN			
Module	Contents	CO	Hours
I	Introduction to Sustainability in the Built Environment and Green Buildings - Environmental impacts of conventional buildings - Sustainability in the built environment - Introduction to life cycle assessment (LCA) and carbon footprint - Concept, definition and need of green buildings – SDGs, COP, NAPCC	CO1	8
II	Overview of Green Building Rating Systems - Evolution of Green Building Rating Systems - Purpose and structure of rating systems - Global overview: LEED (USGBC), BREEAM (UK), CASBEE, Living Building Challenge, etc. - Comparison of rating systems	CO1 CO2	9
III	Indian Green Building Rating Systems - IGBC (Indian Green Building Council): Structure, points, certification levels - GRIHA (Green Rating for Integrated Habitat Assessment): Categories, evaluation process - ECBC (Energy Conservation Building Code): Overview and compliance - Emerging trends and future directions	CO1 CO2	9
IV	Evaluation Parameters and Tools - Site planning and sustainable site development - Energy efficiency and renewable energy integration - Water conservation and wastewater treatment - Sustainable materials and resources - Indoor environmental quality (IEQ) - Relevant Software/tools	CO3 CO4	10
V	Case Studies and Certification Process - Review of certified buildings (LEED/IGBC/GRIHA) - Green building project documentation - Simulation and performance metrics	CO4 CO5	12

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	Economic analysis: Cost vs. benefit Project Work: Students form teams to assess a building (existing or proposed); Apply a chosen rating system to evaluate building sustainability; Present findings, recommendations, strategies and potential improvements		
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Reference Books

- Kubba, S. (2016). *Handbook of green building design and construction: LEED, BREEAM, and Green Globes* (2nd ed.). Butterworth-Heinemann
- Bauer, M., Möslle, P., & Schwarz, M. (2010). *Green building: Guidebook for sustainable architecture*. Springer
- Kibert, C. J. (2016). *Sustainable construction: Green building design and delivery* (4th ed.). John Wiley & Sons.
- Ching, F. D. K., & Shapiro, I. M. (2014). *Green building illustrated*. Wiley
- U.S. Green Building Council. (2019). *LEED v4 for building design and construction*. U.S. Green Building Council.
- BRE Global. (2016). *BREEAM UK new construction: Technical manual*. Building Research Establishment (BRE)
- Japan Sustainable Building Consortium. (2014). *CASBEE for new construction: Technical manual 2014 edition*. Institute for Building Environment and Energy Conservation (IBEC)
- International Living Future Institute. (2020). *Living building challenge 4.0: A visionary path to a regenerative future*. International Living Future Institute
- GRIHA Council. (2010). *GRIHA manual: Volumes 1–5*. The Energy and Resources Institute (TERI)
- The Energy and Resources Institute. (2004). *Sustainable building design manual: Volume 1 & 2*. TERI Press
- IGBC & GRIHA Official Websites and Certification Guides

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EED002	Sustainable Architecture	2-1-0	3	2025
Course Objectives	The course intends to develop a foundational understanding of sustainable architecture to foster an interdisciplinary approach towards creating sustainable and responsive built environments. By the end of the course, students shall be equipped with the knowledge to contribute to sustainable architectural practices in their respective fields.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome			Learning level
CO 1	Understand the basic concepts of sustainability in architecture.			Understand
CO 2	Apply climate responsive strategies to improve building comfort.			Apply and Analyze
CO 3	Identify alternative building materials and assess real-world examples of sustainable architecture in different contexts.			Analyze and Evaluate
CO 4	Analyze and appraise the role of multiple disciplines in achieving sustainability in built environments.			Analyze and Evaluate
CO 5	Evaluate and integrate sustainable architectural practices as a multidisciplinary collaboration through case examples.			Evaluate and Create
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100 Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 2 marks and will be compulsory. Part B shall contain 2				

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questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 10 marks and can have sub questions.

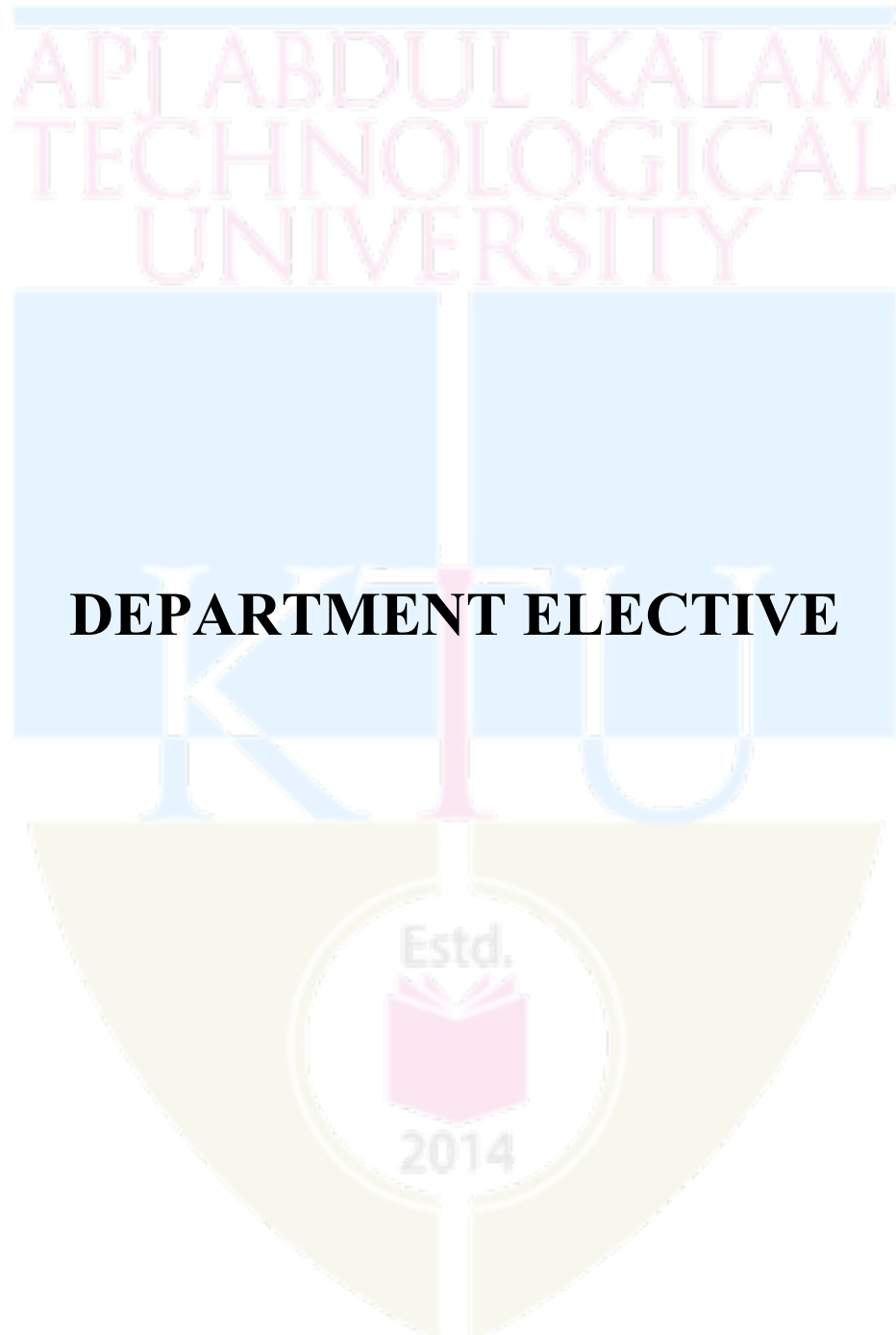
Module	Contents	CO	Hours
I	Introduction to Sustainable Architecture: Introduction to sustainability: Fundamental concepts and definitions. Sustainable architecture in the broader framework of sustainable development, historical development of sustainable practices in architecture. Environmental impacts of the built environment in a global context, relevance of UN Sustainable Development Goals (SDGs)	CO 1	8
II	Characteristics of Sustainable Architecture: Concepts of building comfort: thermal, visual and acoustic Fundamentals of climatology and passive design strategies Life cycle analysis: Cradle to grave, cradle to cradle Sustainability across different building phases	CO 2	8
III	Energy and Carbon Management in buildings Indicators of green buildings Key terminology: Energy efficiency, carbon neutral, carbon positive, Net zero buildings, Carbon footprint, Carbon offset, Carbon credits, Carbon sink, Carbon sequestration, Decarbonization. Urban and rural Environmental issues in building design. Urban Heat Island effect: cause and mitigation strategies, Overview of Green Building Rating systems and examples.	CO 2 CO 3	10
IV	Sustainable Materials and Construction: Overview of conventional building materials and techniques. Introduction to alternative materials and sustainable construction methods. Emerging trends and role of professionals from allied disciplines in sustainable construction with case examples.	CO 3 CO 5	10
V	Sustainable Practices in Architecture: Fundamental lessons from vernacular architecture on sustainable practices Contemporary sustainable practices with relevant Global and Regional Case studies.	CO 4 CO 5	12

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Note: Students shall be encouraged to explore the latest terminologies and advancements in sustainable practices across the globe.

Reference Books/ Readings:

1. Ching, F. D. K., & Shapiro, I. M. (2020). *Green building illustrated* (2nd ed.). Wiley.
2. Mumovic, D., & Santamouris, M. (Eds.). (2018). *A handbook of sustainable building design and engineering: An integrated approach to energy, health and operational performance* (2nd ed.). Routledge.
3. Attmann, O. (2010). *Green architecture: Advanced technologies and materials*. McGraw-Hill.
4. Edwards, B., & Naboni, E. (2013). *Green buildings pay: Design, productivity and ecology* (3rd ed.). Routledge.
5. Maclay, B. (2014). *The new net zero: Leading-edge design and construction of homes and buildings for a renewable energy future*. Chelsea Green Publishing.
6. Merlino, K. R. (2016). *Building reuse: Sustainability, preservation, and the value of design*. University of Washington Press.
7. Cardelus, C. (2018). *Green architecture today*. Braun Publishing.
8. Jodidio, P. (2013). *Green architecture now!*. Taschen.
9. Bolchover, J. (Ed.). (2012). *Vitamin Green*. Phaidon Press.
10. Brown, G. Z., & DeKay, M. (2014). *Sun, wind, and light: Architectural design strategies* (3rd ed.). Wiley.



DEPARTMENT ELECTIVE

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR001	Urban Conservation and regeneration	1-1-0	2	2025
Course Objectives	This course aims to provide a foundational understanding of heritage and conservation principles, including the ethical and cultural dimensions of preserving built environments. It enables students to analyze urban heritage, historic settlements, and cultural landscapes, and equips them with skills in documentation, assessment, and planning of conservation interventions. The course also emphasizes the evaluation of legal frameworks, policies, and institutional roles, while encouraging critical reflection through case studies to develop context-sensitive conservation and regeneration strategies.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand the core concepts, ethics, and evolution of urban conservation and heritage management.	Remembering, Understanding		
CO 2	Analyze typologies and character of historic settlements, urban heritage, and cultural landscapes.	Analyzing		
CO 3	Apply techniques of heritage documentation, assessment, and planning in urban conservation practice.	Applying		
CO 4	Evaluate tools, legal frameworks, and institutional roles in conservation and regeneration processes.	Evaluating		
CO 5	Critically assess case studies and propose context-sensitive conservation or regeneration strategies.	Analyzing and Evaluating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				

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End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 5 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 7 marks and can have sub questions.

Module	Contents	CO	Hours
I	Fundamentals of Conservation Definitions: Tangible and intangible heritage built and cultural heritage, conservation vs. preservation: ethics, principles, degrees of intervention. Distinction between architectural and urban conservation. Concepts of authenticity, integrity, and cultural landscapes. International and Indian conservation paradigms.	CO1, CO2	6
II	Urban Heritage and Historic Landscapes Components of urban heritage: morphology, public spaces, uses, cultural practices and traditions Historic cities and precincts: Typologies Concept of historic precincts and cities, Cultural Landscape, Historic Urban Landscape (HUL), charters pertaining to them and case examples. World Heritage Cities: Case examples and criteria Historic cities: Indian examples, Case studies to identify components and Issues in Historic cities/cores.	CO1, CO2, CO5	6
III	Conservation Process and Practices Urban conservation process from archival research to management plans. Tools and Techniques: Inventories, photogrammetry, mapping, visual surveys, Heritage Impact Assessments (HIAs) Case examples of interventions in Historic cores: Adaptive reuse, restoration, rehabilitation, infill in historic areas. Role of public participation, and heritage-led regeneration.	CO2, CO3, CO5	6

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IV	Tools, Legislations, and Institutional Frameworks Legal instruments: AMASR Act, CRZ, DCRs, Wetlands Rules, Masterplans: Special zones National, state institutions and NGOs: ASI, INTACH, UAC, MoHUA Conservation schemes, funding models, and urban policy integration - AMRUT, HRIDAY, JNNURM, CSR in Heritage conservation Role of ULBs, state departments, community groups Heritage and Tourism: Prospects and problems	CO1, CO4, CO5	7
V	Urban Regeneration and Case Studies Concepts: Redevelopment, renewal, rejuvenation, reconstruction Triggers for regeneration: degradation, obsolescence, socio-economic shifts Planning and implementation strategies: phasing, financing, legal tools Comparative regeneration models: East and West Case studies: Global and National.	CO3, CO4, CO5	7

Reference Books

- Feilden, B. M. (2003). *Conservation of historic buildings* (3rd ed.). Routledge.
- Glendinning, M. (2013). *The conservation movement: A history of architectural preservation*. Routledge.
- Menon, A. G. K. (Ed.). (n.d.). *Conservation of immovable sites*. INTACH Publication.
- Hollis, M. (2000). *Surveying buildings*. RICS Business Services Limited.
- Silva, K. D., & Chapagain, N. K. (2013). *Asian heritage management: Contexts, concerns and prospects*. Routledge.
(Please confirm publisher and year; Routledge is common for this title.)
- INTACH. (n.d.). *Heritage and conservation: An introductory handbook*. INTACH.
- Banga, I. (2005). *The city in Indian history*. Manohar Publishers and Distributors.
- ICOMOS, & UNESCO. (n.d.). *Charters and guidelines*.
(If available, specify edition or publication year.)
- Cody, J., & Siravo, F. (2019). *Historic cities: Issues in urban conservation*. Getty Conservation Institute.
- Ministry of Housing and Urban Affairs, Government of India. (n.d.). *HRIDAY reflections: A monograph on the heritage city development and augmentation Yojana*.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR002	Environmental Impact Assessment	1-1-0	2	2025
Course Objectives	This course aims to determine and forecast the impact of a project on the biological, geophysical, and chemical environment, as well as human health, and suggest suitable policies, initiatives, and operational practices to mitigate this impact.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Identify both favourable and unfavourable potential environmental effects of a given project. Analyze how the "short-term" and the “long-term" time frames are used to identify these effects.	Remember, and Analyze		
CO 2	Formulate strategies that, when implemented, will mitigate or offset the adverse environmental impacts of a project that causes minimal environmental degradation.	Create		
CO 3	To assess how well environmental protection regulations are being implemented and how beneficial they are.	Evaluate		
CO 4	To prepare EIA reports with a collaborative approach	Create		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 5 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 7 marks and can have sub questions.				

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Module	Contents	CO	Hours
I	Fundamental Approach to Environmental Impact Assessment: Definition, principles, and objectives, Historical evolution Legislation and Authorities concerning EIA: National & International EIA Methodologies: checklists, matrices, and modelling	CO1	6
II	Prediction and Assessment of Impacts on Soil and Groundwater Environment, Surface Water Environment, & Air Environment. Case Studies and Cross-Media Effects	CO1 CO3	6
III	Prediction and Assessment of Impacts on Biological Environment, Noise on the Environment, Socio-Economic and Human Health Impacts, Community engagement.	CO3	6
IV	Environmental Risk Assessment (ERA) and Risk Management in EIA, Application for Remote Sensing and GIS for EIA, Quantitative and Qualitative Risk Methods	CO3	7
V	EIA Case Studies review and appraisal, EIA Classroom Project, Preparation of an EIA report, Presentation of EIA reports.	CO2 CO4	7

Reference Books

1. Lohani, B. N., et al. (1997). *Environmental impact assessment for developing countries in Asia*. Asian Development Bank.
2. Raman, N. S., et al. (2014). *Environmental impact assessment*. TechSar Pvt Ltd.
3. Glasson, J., Therivel, R., & Chadwick, A. (2012). *Introduction to environmental impact assessment*. Taylor & Francis Ltd.
4. Wathern, P. (2015). *Environmental impact assessment: Theory and practice*. Routledge.
5. Satsangi, A., & Sharma, A. (2015). *Environmental impact assessment (EIA) and disaster management*. Rajat Publications.
6. Srivastava, A. K. (2012). *Environmental impact assessment: Theory and practice*. APH Publishing Corporation.
7. Gilpin, A. (1994). *Environmental impact assessment: Cutting edge for the 21st century*. Cambridge University Press.
8. Das, S. (2024). *Environmental planning and impact assessment: Strategies for sustainable management*. Global Net Publication.

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9. The World Bank. (1991). *The environmental assessment sourcebook* (Vols. I–III). The World Bank.
10. Anjaneyulu, Y., & Manickam, V. (2020). *Environmental impact assessment methodologies* (3rd ed.). BS Publications.
11. Reddy, P. J. (2016). *Environmental impact assessment theory & practice*. BS Publications.



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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR003	Urban Sociology	1-1-0	2	2025
Course Objectives	This course introduces key concepts of Urban Sociology and gives special attention to the emergence and growth of Urban Sociology, social structure in Urban society and challenges in urban governance. This course will provide an exposure to key theoretical perspectives for understanding urban phenomena in historical and contemporary contexts. With case studies from India and other parts of the world, this course will help students understand and relate to the complexities of urban living. This course seeks to evolve critical thinking and policy perspectives on sustainable urban development.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Understand and analyze fundamental urban sociological concepts, its emergence and growth and understand types of cities with special emphasis on Indian cities.	Understanding		
CO 2	Critically analyze and apply theories of Urban sociology and theories of Urbanization.	Applying		
CO 3	Understand Urban Social Structure, inequalities and polarization, social area analysis and factorial ecology.	Understanding		
CO 4	Apply best practices of Urban governance to address Urban social problems	Applying		
CO 5	Analyze and understand social psychology and social geography and address gender and equity concerns in urban development.	Understanding & applying		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				

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End Semester Examination pattern:

There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 5 marks and will be compulsory. Part B shall contain 2 questions from each module, out of which one shall be attempted by the student. The questions in this part will carry 7 marks and can have sub questions.

Module	Contents	CO	Hours
I	Introduction to Urban Sociology Urban Sociology: meaning, definition, nature and scope. Basic concepts: Urbanization, Urbanism, Urbanity, Rural-Urban Fringe, Conurbation, Suburbanization, Peri urbanization, Ghettoization, Urban neighbourhoods. Brief history of Urban Sociology: Origin and development of urban societies. Types of cities: Preindustrial City, Industrial City and Post-Industrial city; Examples. Indian cities: demographic, economic and ecological characteristics, case examples.	CO1	6
II	Theories of Urban Sociology and Urbanization Classical theories of City: Theories of Max Weber; George Simmel. The Chicago School of Sociology and its Critics: Theories of Robert Park, Burgess & Louis Wirth. Urban Anomie Theory: Emile Durkheim, Robert K. Merton. The New Urban Sociology: Henri Lefebvre, David Harvey, Manuel Castells. Theories of Urbanization: Theory of spatial disparity, Migration theories, Bid Rent theory, Modern theory of urbanization. Application of theories in contemporary urban contexts.	CO2	7
III	Urban Social Structure and Inequalities Urban Social Structure: definition, types of urban social structure, case examples Community in urban setting: identity & difference, Urban Culture & Urban activities Inequalities and Polarisation: patterns and types of inequality; Group Polarisation, Apartheid, Ethnic segregation of space, Gentrification, Marginalization, Gender Inequality. Social Area Analysis and Factorial Ecology, Case example.	CO3	7

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IV	Urban Social Problems and Urban Governance Urban Social Problems: Crime and Juvenile delinquency, Fear of Crime, Slum & Health, Poverty & Unemployment, Immigration, Beggary, Problems of women and children, Urban Violence, Terrorism in India. Solutions to urban social problems. Best Practices. Urban Governance: Participatory & Inclusive governance, Challenges of urban governance in Indian context, Measures to strengthen local governance to realize sustainable urban development.	CO4	6
V	Social Psychology and Social Geography Social Psychology: definition, basic concepts. Nature and scope of social psychology in Indian context. Theories of social psychology & Examples of Social psychological experiments. Social geography: Models of social geography, European perspectives of social geography, Image of the city and Mental mapping, examples. Planners and urban designers' role in empowering the image of cities. Ecofeminism, Production of space geography of gender in city, Case examples.	CO5	7

Reference Books

1. Flanagan, William G. *Urban Sociology: Images and Structures*, 5th Edition, Rowman and Littlefield Publishers Inc., New York, 2010.
2. Sharma K.R., *Urban Sociology*, New Delhi: Atlantic Publishers, 1997.
3. Bhushan, Vidya & D.R. Sachdeva. *An Introduction to Sociology*, Kutab Mahal distributors, 2014.
4. Burgess, Ernest W., and Robert E.Park, *The City*, Chicago, IL: University of Chicago Press, 1984.
5. M.Francis Abraham. *Modern Sociological Theory- An Introduction*, Delhi Oxford University Press, 1995.
6. Richard T.Schaeffer, Robert P.Lamm, *Sociology*, International edition, McGraw-Hill Inc., 2001.
7. James W. Vander Zanden. *The Social Experience-An Introduction to Sociology*, International Edition, McGraw Hill Publishing Co.,1990.
8. Michael, S.M. *Culture and Urbanization*, Delhi: Inter-India Publications.,1989.
9. Murray Lee, Gabe Mythen, *The Routledge International Handbook on Fear of Crime*, Taylor & Francis Group, 2017.
10. Carter, H., *The Study of Urban Geography*, Edward A. Old London, 1972.
11. Sibnath Deb, Anjali Gireesan, Pooja Prabhavalkar, Shayana Deb, *Social Psychology- Theories and Applications*, Routledge, New York, 2023.

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COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
252EAR004	Contemporary Pedagogy in Architectural Education	1-1-0	2	2025
Course Objectives	This course explores the theories, methods, and practices of architectural education, with a focus on pedagogy as it applies to design studios, theory classrooms, and interdisciplinary learning environments. It equips future educators, academic professionals, and researchers with the tools to critically engage with curriculum design, studio instruction, and educational innovation in architecture.			
Course Outcomes: After the completion of the course, the students will be able to:				
CO	Description of Course Outcome	Learning level		
CO 1	Analyze historical and theoretical foundations of architectural pedagogy.	Analyzing		
CO 2	Design inclusive and effective studio-based and classroom-based learning environments.	Creating		
CO 3	Create curriculum structures aligned with contemporary educational standards and design practices.	Creating		
CO 4	Apply fair and meaningful assessment practices in design evaluation	Applying		
CO 5	Reflect on their own teaching philosophies and develop strategies for academic and institutional leadership.	Evaluating and Creating		
Mark Distribution: Continuous internal evaluation - 40 End semester examination - 60 (Duration - 2.5 hours) Total Marks - 100				
Continuous Internal Evaluation pattern: Assignment/ Quiz/ Course project - 25 marks Internal test - 15 marks				
End Semester Examination pattern: There will be two parts; Part A and Part B. Part A shall consist of 5 questions, one from each module. Each question shall carry 5 marks and will be compulsory. Part B shall contain 2 questions.				

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from each module, out of which one shall be attempted by the student. The questions in this part will carry 7 marks and can have sub questions.

Module	Contents	CO	Hours
I	Foundations of Architectural Pedagogy – - History and evolution of Architectural Education - Pedagogical theories & philosophies - Bloom’s Taxonomy of Educational Objectives - Role of the educator and the learner in architectural studios - Critique culture and peer learning	CO1, CO2	6
II	Teaching Models and Experiential Learning – - Models of Teaching – Advanced organizer model, concept attainment model, simulated-based learning, Synectics model - Experiential Learning - Relevance of these models in architectural education	CO2 CO3	6
III	Techniques for Teaching and Learning – - Maxims of teaching and their application in architecture - Teaching tools: Concept mapping and creation of Concept maps - Micro-teaching skills - Creative thinking techniques	CO2, CO3, CO5	6
IV	Assessment and Feedback in Design Education - - Evaluating Design, Rubrics - Assessment methods in education - Peer review, self-assessment, critique reflection - Role of Inventory in Education - Aligning feedback with learning outcomes	CO3, CO4	7
V	Emerging Trends in Education, Pedagogical Research and Supervision – - Future of Architectural Education – AI, Moodle, Sustainability, Hybrid learning - Indian and Global perspectives – NEP 2020, SDGs, International Models - NAAC and NBA Accreditations - Scholarship of Teaching and Learning (SoTL) - Pedagogical research methods – qualitative, action, case study	CO5, CO1, CO3	7

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	• Basics of mentoring and academic supervision • Ethics and roles in supervision		
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Textbooks:

1. Salama, A. M. (2015). *Spatial Design Education: New Directions for Pedagogy in Architecture and Beyond*. Ashgate.
2. Biggs, J. & Tang, C. (2011). *Teaching for Quality Learning at University*. McGraw-Hill.
3. Joyce, B., Weil, M., & Calhoun, E. (2015). *Models of Teaching*. Pearson.
4. Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action*. Basic Books.

References Books and Reports:

1. Ramsden, P. (2003). *Learning to Teach in Higher Education*. Routledge.
2. Lawson, B. (2005). *How Designers Think: The Design Process Demystified*. Architectural Press.
3. Brookfield, S. D. (2017). *Becoming a Critically Reflective Teacher*. Jossey-Bass.
4. Gordon, W. J. J. (1961). *Synectics: The Development of Creative Capacity*. Harper & Row.
5. Ministry of Education, India. (2020). *National Education Policy (NEP 2020)*.
6. UNESCO-UIA. *Charter for Architectural Education*.
7. Kolb, D. A. (1984). *Experiential Learning: Experience as the Source of Learning and Development*. Prentice Hall.
8. Cross, N. (2011). *Design Thinking: Understanding How Designers Think and Work*. Berg.

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Master of Architecture (Environmental Design) – Curriculum

COURSE CODE	COURSE NAME	L-T-S	CREDITS	YEAR OF INTRODUCTION
253MED001	Elective I - MOOC	0-0-0-0	2	2025

Massive Open Online Course (MOOC) introduced and made mandatory for M Arch Environmental Design degree course is meant to provide opportunities for learners to get wider exposure, through flexible learning. The courses offered in varied topics are handled by top resource people, which are easily accessible and affordable, transcending geographical barriers.

Instructions for MOOC course

1. Students should choose one MOOC Certificate course from AICTE / SWAYAM or NPTEL. The MOOC can either be from the approved list given by the university or by the concerned college.
2. The MOOC should have a **minimum duration of 8 weeks**, and the content of the syllabus should be enough for **at least 40 hours of teaching**.
3. The course should include a **proctored/ offline examination** at the end of the course.
4. For securing the credits, the student is required to **complete the assessment** of the course and to provide the **certificate** of the course.
5. The fees (if any) for the registration and/or assessment of the MOOC course must be borne by the student only.
6. Even if the student completes more than one MOOC course, even then **2 credits** shall be considered for this subject.
7. The students can do the MOOC according to their convenience but should **complete it by the third semester** of the M Arch (ED) course.
8. Passing grades must be obtained before the last working day of the 3rd semester for the award of credits without which the student shall not be eligible for award of M. Arch Degree.
9. The registration details for the MOOC as well as the completion certificate shall be submitted duly with the MOOC Coordinator of the institute.
10. Two credits are given for this on a pass or fail basis and it is mandatory for getting M Arch (ED) degree. As no grade is given, this will not be used for CGPA calculation.

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The suggested broad areas are defined as follows:

- **Sustainable Architecture Design** (Passive and Active Design Strategies, Net Zero Energy Building, Life Cycle Assessment (LCA), Low Carbon Building Design, etc.)
- **Climate Responsive Design** (Design for Specific Climatic Region, Bioclimatic Architecture, Landscape Design & Site Planning, etc.)
- **Green Building Certifications** (LEED, GRIHA & and certification programs)
- **Environmental Management** (EIA, Resource and Waste management, Renewable energy, etc.).
- **Building Science & Applied Engineering** / Building Services / Building Technology (Building Materials and Performance, Acoustics, Lighting Design, etc.)
- **Ecological Urbanism** (Integrating ecosystem into city planning, Urban metabolism and Circular Design, Sustainable Urban Design, etc.)
- **Analytical & Technical Tools for Environmental Design** (Environmental Simulation Tools, GIS, Building Performance Analytics, etc.)
- **Specialized & Emerging Topics** (Carbon Neutral Architecture, Architectural Biomimicry, Environmental policies for Architects, Resilient Design for Climate Adaptation, Digital Sustainability - Using AI and parametric design for eco-optimization, etc.)

* Students can also choose topics / courses relevant to Environmental Design offered by agencies, namely AICTE / NPTEL / SWAYAM with the recommendation of the Board of Studies (at least 70% of the course content shall be matched with the area/stream of study) and approval by the Academic Council.

*The Course shall not be considered if its content has more than 50% of overlap with a core/elective course in the concerned discipline or with an open elective.

Approved MOOCs providers:

SWAYAM MHRD <https://swayam.gov.in>

NPTEL IITs, IISc <https://onlinecourses.nptel.ac.in>